Grade 7 Computing

Strand 1: Introduction to Computing

Lesson 3 — Technology in the Community

Codes: B7.1.2.1.1 – B7.1.2.1.6 **Format:** Theory Questions and Answers

Preamble (pre-lesson)

Technology is not only in the computer lab. It is in the market, the hospital, the bank, and on the road.

This lesson answers questions about:

1. **Technology tools in the community.**
2. **Using some of those tools.**
3. **Benefits of technology for learning.**
4. **Negative impacts of computers.**
5. **Environmentally responsible practices.**
6. **Making new things from old computer parts.**

Words to know first:

- **Technology tool** — any device or system that helps people do work.
- **Community** — the people and places around where you live.
- **Impact** — the effect something has, good or bad.
- **Environmentally responsible** — acting in a way that protects nature.
- **Dispose** — get rid of something properly.
- **E-waste** — old or broken electronics.
- **Upcycle** — make something useful from old items.

A. Technology tools in the community

Q1. What is a technology tool? **Ans.** A technology tool is any device or system that helps people do work.

Q2. Mention five technology tools found in your community. **Ans.**

- Mobile phone
- ATM
- Point-of-sale (POS) card reader
- Digital weighing scale
- Traffic light

Q3. What does a mobile phone do in the community? **Ans.** A mobile phone helps people communicate and use services like mobile money.

Q4. What does an ATM do? **Ans.** An ATM gives people cash and shows account balances.

Q5. What does a point-of-sale (POS) card reader do? **Ans.** A POS card reader accepts card and mobile money payments.

Q6. What does a digital weighing scale do? **Ans.** A digital weighing scale measures the mass of goods accurately.

Q7. Where do people use barcode scanners in the community? **Ans.** People use barcode scanners in shops and supermarkets.

Q8. What does a traffic light do? **Ans.** A traffic light controls the flow of vehicles and people at a road junction.

Q9. Where do people use digital scanners and ultrasound machines? **Ans.** People use them in hospitals and clinics.

Q10. Where do people use printers and photocopiers in the community? **Ans.** People use them in schools, offices, and business centres.

B. Using technology tools

Q11. How do you make a call on a mobile phone? **Ans.** You open the Phone app, dial the number, and press call.

Q12. How do you print a document from a computer? **Ans.** You open the document, choose File, then choose Print, and select the printer.

Q13. How do you read a digital weighing scale? **Ans.** You read the number shown on the display together with its unit.

Q14. What does a POS card reader confirm after a payment? **Ans.** It confirms that the payment has been received.

Q15. How does a barcode scanner help a shop? **Ans.** It helps the shop read prices and record stock quickly.

C. Benefits of technology for learning

Q16. Mention four benefits of technology for learning. **Ans.**

- Faster access to information
- Better explanations using videos and pictures
- Learning from home
- Quick feedback from quizzes

Q17. How does technology make learning faster? **Ans.** Technology gives access to information in seconds instead of searching through many books.

Q18. How do videos help explanations? **Ans.** Videos show processes step by step in a way that is easy to follow.

Q19. How does technology help students learn from home? **Ans.** Students can use the radio, television, and internet to study outside the classroom.

Q20. How do quizzes give feedback? **Ans.** Quizzes mark answers immediately and show the learner what to improve.

Q21. How does technology help collaboration? **Ans.** Technology lets a group work on shared documents at the same time.

Q22. How does technology help keep records? **Ans.** Technology stores notes safely so they can be found again later.

D. Negative impacts of computers

Q23. Mention four negative impacts of computers. **Ans.**

- Eye strain and back pain
- Distraction from study
- Cyberbullying
- E-waste

Q24. What is cyberbullying? **Ans.** Cyberbullying is sending hurtful messages or images to someone online.

Q25. What is e-waste? **Ans.** E-waste is old or broken electronic equipment that has been thrown away.

Q26. How can computers distract students from study? **Ans.** Games and social media can take up time that should be used for learning.

Q27. What is a privacy risk when using computers? **Ans.** A privacy risk is when personal information is shared with strangers online.

Q28. Why is too much time on a computer harmful? **Ans.** Too much time on a computer can cause addiction, poor sleep, and health problems.

E. Environmentally responsible practices

Q29. What does it mean to be environmentally responsible? **Ans.** It means acting in a way that protects nature and does not harm the land.

Q30. Mention three environmentally responsible practices for computers. **Ans.**

- Reduce the number of devices you buy
- Reuse working equipment
- Recycle e-waste at proper collection points

Q31. What does "reduce" mean in handling e-waste? **Ans.** Reduce means buying only what you need and choosing devices that last.

Q32. What does "reuse" mean in handling e-waste? **Ans.** Reuse means repairing and passing on equipment that still works.

Q33. What does "recycle" mean in handling e-waste? **Ans.** Recycle means sending old electronics to proper collection points instead of dumping them.

Q34. Why should you not burn old electronics? **Ans.** Burning electronics produces toxic smoke that harms people and the land.

Q35. Why should you not dump electronics in waterways? **Ans.** Dumping electronics in waterways pollutes the water and harms living things.

Q36. How do you save energy when using computers? **Ans.** You shut down the computer and unplug chargers when they are not in use.

F. Upcycling computer parts

Q37. What is upcycling? **Ans.** Upcycling is making something useful from old or waste items.

Q38. Give one example of a project made from disposed computer parts. **Ans.** A keyboard-key keychain is one example of a project made from disposed computer parts.

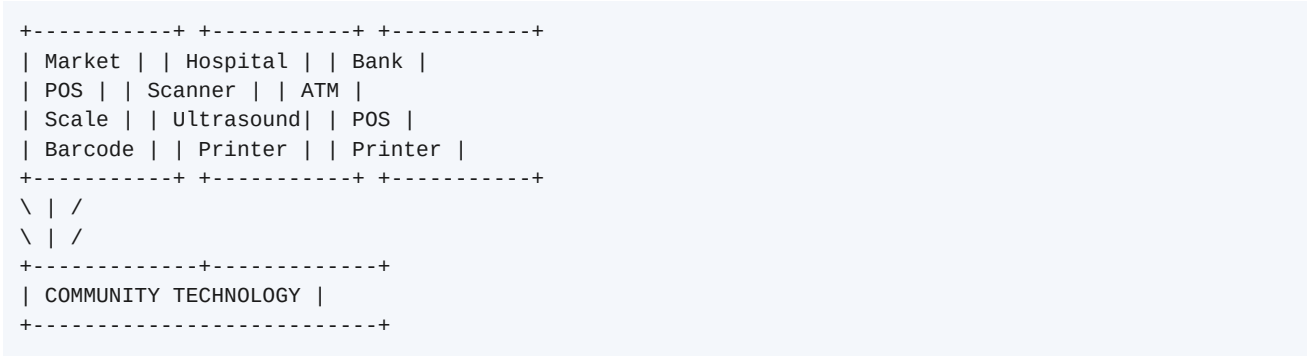
Q39. Give another example of a project made from old computer parts. **Ans.** A USB pencil holder made from a dead drive casing is another example.

Q40. Which computer part must you never open? **Ans.** You must never open a power supply unit or a CRT monitor.

Q41. Why must you never open a power supply unit? **Ans.** A power supply unit can keep a dangerous electric charge even when it is unplugged.

Q42. How do you make a robot sculpture from old parts? **Ans.** You join old mice, cables, and cases together into a standing shape.

Text diagram — community technology map



Text diagram — the 3 R's of e-waste

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REDUCE --> buy less, choose devices that last
|
v
REUSE --> repair and pass on working equipment
|
v
RECYCLE --> send e-waste to proper collection points
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Exercises (8 sections × 5 questions)

Exercise 1 — Tools in the community

1. What is a technology tool?
2. Mention five technology tools found in your community.
3. What does an ATM do?
4. What does a digital weighing scale do?
5. **[Review]** What is an input device?

Answers

1. A technology tool is any device or system that helps people do work.
2. Mobile phone, ATM, POS card reader, digital weighing scale, traffic light.
3. An ATM gives people cash and shows account balances.
4. A digital weighing scale measures the mass of goods accurately.
5. An input device is a tool that puts data into a computer.

Exercise 2 — Using tools

1. How do you make a call on a mobile phone?
2. How do you print a document from a computer?
3. What does a POS card reader confirm?
4. How do you read a digital weighing scale?
5. **[Review]** What does a barcode scanner do?

Answers

1. You open the Phone app, dial the number, and press call.
2. You open the document, choose File, then choose Print, and select the printer.
3. It confirms that the payment has been received.
4. You read the number shown on the display together with its unit.
5. A barcode scanner reads the striped code printed on a product.

Exercise 3 — Benefits for learning

1. Mention four benefits of technology for learning.
2. How does technology make learning faster?
3. How do videos help explanations?
4. How do quizzes give feedback?
5. **[Review]** What is the 20-20-20 rule?

Answers

1. Faster access to information; better explanations; learning from home; quick feedback.
2. Technology gives access to information in seconds instead of searching through many books.
3. Videos show processes step by step in a way that is easy to follow.
4. Quizzes mark answers immediately and show the learner what to improve.
5. The 20-20-20 rule says that every 20 minutes you should look at something 20 feet away for 20 seconds.

Exercise 4 — Negative impacts

1. Mention four negative impacts of computers.
2. What is cyberbullying?
3. What is e-waste?
4. How can computers distract students from study?
5. **[Review]** What is posture?

Answers

1. Eye strain and back pain; distraction from study; cyberbullying; e-waste.
2. Cyberbullying is sending hurtful messages or images to someone online.
3. E-waste is old or broken electronic equipment that has been thrown away.
4. Games and social media can take up time that should be used for learning.
5. Posture is the way you hold your body when you sit or stand.

Exercise 5 — Environmentally responsible practices

1. What does it mean to be environmentally responsible?
2. Mention three environmentally responsible practices for computers.
3. What does "reduce" mean in handling e-waste?
4. Why should you not burn old electronics?
5. **[Review]** What is a surge protector?

Answers

1. It means acting in a way that protects nature and does not harm the land.
2. Reduce the number of devices; reuse working equipment; recycle e-waste.
3. Reduce means buying only what you need and choosing devices that last.
4. Burning electronics produces toxic smoke that harms people and the land.
5. A surge protector is a device that protects equipment from sudden spikes of electricity.

Exercise 6 — Energy and disposal

1. How do you save energy when using computers?
2. Why should you not dump electronics in waterways?
3. What does "recycle" mean in handling e-waste?
4. How does unplugging chargers help?
5. **[Review]** What is the difference between an HDD and an SSD?

Answers

1. You shut down the computer and unplug chargers when they are not in use.
2. Dumping electronics in waterways pollutes the water and harms living things.
3. Recycle means sending old electronics to proper collection points instead of dumping them.
4. Unplugging chargers saves electricity and reduces the risk of fire.
5. An HDD uses spinning disks while an SSD has no moving parts.

Exercise 7 — Upcycling project

1. What is upcycling?
2. Give one example of a project made from disposed computer parts.
3. Which computer part must you never open?
4. Why must you never open a power supply unit?
5. **[Review]** What is a folder?

Answers

1. Upcycling is making something useful from old or waste items.
2. A keyboard-key keychain.
3. You must never open a power supply unit or a CRT monitor.

4. A power supply unit can keep a dangerous electric charge even when it is unplugged.
5. A folder is a container that holds files and other folders.

Exercise 8 — Scenario and discussion

1. A friend spends six hours daily on games and cannot study. What advice do you give?
2. Your community dumps phones in the river. What is the impact and what should they do instead?
3. Write one sentence you can put on a poster about e-waste.
4. **[Review]** What does the Power option in Windows do?
5. **[Review]** Mention four output devices.

Answers

1. Set a daily time limit, use the computer first for learning, and take regular breaks.
2. The river becomes polluted; they should take the phones to an e-waste collection point instead.
3. Bin the habit, not the device; recycle your e-waste.
4. The Power option lets you Sleep, Shut down, or Restart the computer.
5. Monitor, printer, speakers, projector.

Summary table

Term	One-line meaning
Technology tool	a device that helps people do work
Impact	the effect something has, good or bad
E-waste	old or broken electronics
Reduce / Reuse / Recycle	the 3 R's of e-waste
Upcycling	making something useful from old items

NaCCA CCP B7.1.2 — Theory Questions and Answers format. Plain simple English. British notation.